Bridging the Menopause Information Gap: A Peer-Reviewed Multimodal RAG Framework with Hybrid Retrieval

Sara El-Ateif*† and Soufiane Elassri†
*Laboratory of Computer Systems Engineering, Mathematics and Applications,
Polydisciplinary Faculty of Taroudant, Ibn Zohr University, B.P. 8106, Agadir, Morocco
†National Higher School of AI and Data Science (ENSIASD),
Ibn Zohr University, Taroudant, Morocco
(s.el-ateif@, soufiane.elassri.60@edu.)uiz.ac.ma

***Abstract*—The current study addresses the significant challenge of accessing reliable, personalized medical information for women experiencing menopause. While online data is often contradictory, specialized consultations can be cost-prohibitive and difficult to access, creating a critical need for structured, scientifically validated knowledge. To this end, we propose an intelligent system based on a Multimodal Retrieval-Augmented Generation (MRAG) architecture designed to centralize and personalize access to medical information. Our approach leverages MRAG coupled with a curated corpus of peer-reviewed literature (PLOS ONE) to mitigate the risk of LLM hallucinations. The proposed methodology involves: (1) utilizing the Gemini 2.0 Flash vision model to generate semantic captions for clinical imagery, thereby enabling text-based retrieval of multimodal assets; (2) employing a hybrid search mechanism that combines vector similarity and lexical correspondence (BM25), optimized via cross-encoder reranking and semantic repacking; and (3) using an LLM-based query classifier to route requests between direct generation and full retrieval. Empirical evaluations demonstrated the system's efficacy, achieving a routing accuracy of 96% on the evaluation set. Furthermore, the system yielded a source faithfulness rate of 88% ($\pm$3.1%), a response relevance of 90% ($\pm$2.8%), and a contextual precision of 85% ($\pm$3.6%). A limited preliminary user assessment resulted in a System Usability Scale score of 82/100, indicating the solution's strong potential for digital women's health and personalized medicine designed to supplement, rather than replace, professional medical consultation.**

***Index Terms*—Digital Women's Health, Menopause, Multimodal Indexing, RAG, Information Retrieval.**

I. Introduction

Menopause is a universal physiological transition experienced by women, typically occurring between the ages of 45 and 55. It involves a significant decline in estrogen and progesterone levels, resulting in symptoms that range from vasomotor episodes and sleep disruption to mood instability and cognitive changes [1]. The clinical presentation varies considerably among individuals, which makes generic advice insufficient for personalized care.

Access to reliable online health information regarding menopause has not kept pace with the growing need. Searches for menopause-related symptoms frequently yield a mix of unverified blog posts, forum threads, and commercially motivated content, much of which is contradictory. Furthermore, a significant portion of the global population remains underserved by digital health tools, which are predominantly designed for English-speaking audiences. This lack of access to structured, evidence-based guidance exacerbates existing inequalities in access to specialist care.

Recently, Large Language Models (LLMs) have shown potential in bridging this information gap. However, their tendency to generate plausible-sounding yet factually incorrect information (hallucinations) renders them unsuitable as direct patient-facing tools without rigorous validation mechanisms [3]. Retrieval-Augmented Generation (RAG) [2] was designed precisely to address this. Rather than relying on parametric knowledge alone, the model grounds its answers in documents retrieved at inference time from a controlled knowledge base. This shifts the quality bottleneck from the model to the quality of the corpus, which is exactly where we focus our effort.

We built a specialized assistant that retrieves information from a *Gold-Standard Curated Corpus* of 100 peer-reviewed menopause articles, processes both text and visual content from these documents, and delivers answers through a simple Streamlit interface. This work is best understood as an applied systems integration contribution. The novelty of our work lies not in new retrieval algorithms, but in the principled combination of curated multimodal retrieval, hybrid search, and a complete patient-facing application within a single domain-specific, locally-deployable package. The main technical contributions are:

1) A domain-restricted corpus built from PLOS ONE [5], scraped and validated via an automated Selenium pipeline, prioritizing source quality over scale.
2) A multimodal indexing pipeline that uses Gemini 2.0 Flash to caption clinical figures and graphs, enabling visual content to be retrieved through natural language.
3) A hybrid retrieval engine combining BGE-large dense embeddings with BM25 sparse search, followed by cross-encoder reranking and semantic repacking.
4) An LLM-based query classifier that routes queries to

either direct generation or the full RAG pipeline, with a characterized routing accuracy.

5) A user-facing interface with four tabs—a RAG chatbot, a symptom severity tracker, a personalized PDF report generator, and an educational cards panel—evaluated using the SUS questionnaire.

II. Related Work

A. RAG in Healthcare

RAG [2] has become the default architecture for reducing hallucination in medical question answering. Handy et al. [4] applied this approach to the menopause domain by building a chatbot on top of GPT-4 and reported strong initial metrics for faithfulness (88.6%) and relevance (91.8%). On the broader medical side, Zhu et al. [7] proposed EMERGE, which integrates RAG into EHR-based predictive modeling, and a recent systematic review by Liu et al. [8] found that RAG-augmented systems outperform standard LLMs in diagnostic tasks with an odds ratio of 1.35 (95% CI [1.19–1.53]).

Several task-specific systems are also worth noting. Xu et al. [9] demonstrated that a RAG-based nursing advisory tool for breast cancer patients performed best when the underlying corpus was tightly scoped rather than general. Alkhalaf et al. [10] showed 99.3% faithfulness in medical record summarization by similarly restricting source documents. Zhou et al. [11] explored adapting retrieval strategies to individual patient history—an idea we apply here through the symptom tracker. A Portuguese RAG chatbot for women's reproductive health has also demonstrated that language-specific health assistants are both feasible and in-demand [12].

B. Multimodal and Specialized Architectures

Medical documents are inherently multimodal. For menopause, hormone level charts, anatomical diagrams, and clinical trial figures all carry critical information that text chunks cannot represent. Including visual data alongside text has been proven to provide better results as it uses complementary information offered by each modality. Xia et al. [13] quantified this gap. Their MMed-RAG system showed a 43.8% improvement in factual precision when processing visual data with text. Lahiri et al. [14] applied multimodal RAG to Alzheimer's literature, and Yu et al. [15] used a similar setup for automated radiology report generation.

On the retrieval optimization side, Sohn et al. [16] introduced RAG^2 to filter out low-quality retrieved passages before generation, while Minh et al. [17] added a self-reflection step in Self-BioRAG to improve biomedical reasoning. Knowledge graph integration has also been explored—Liu et al. [23] built MKRAG and Zhao et al. [24] developed MedRAG, both showing gains on complex clinical queries. Zhang et al. [25] proposed BioMedRAG as a more standardized supervised retrieval baseline for biomedical QA.

C. Gap Analysis

What stands out from the literature is that most systems handle either text or images, but rarely both in a tightly integrated pipeline. Hybrid dense–sparse retrieval is frequently described as beneficial but seldom implemented in the medical domain. Women's health in general, and menopause in particular, is almost entirely absent from the RAG literature. And to the best of our knowledge, no existing system combines curated multimodal retrieval with end-user tools like symptom tracking and PDF report export in a single deployable application. Our work attempts to fill all four of these gaps at once.

III. Methodology

Fig. 1 gives an overview of the system. A user query first passes through an LLM classifier that determines whether direct generation is sufficient or whether retrieval is needed (see Section III-C). When retrieval is triggered, the query hits both a dense vector index and a BM25 index built from our curated corpus; results are merged, reranked by a cross-encoder, repacked to remove redundancy, and then passed to the generator. The interface exposes this through four tabs: a RAG chatbot, a symptom tracker, a PDF report panel, and an educational cards panel.

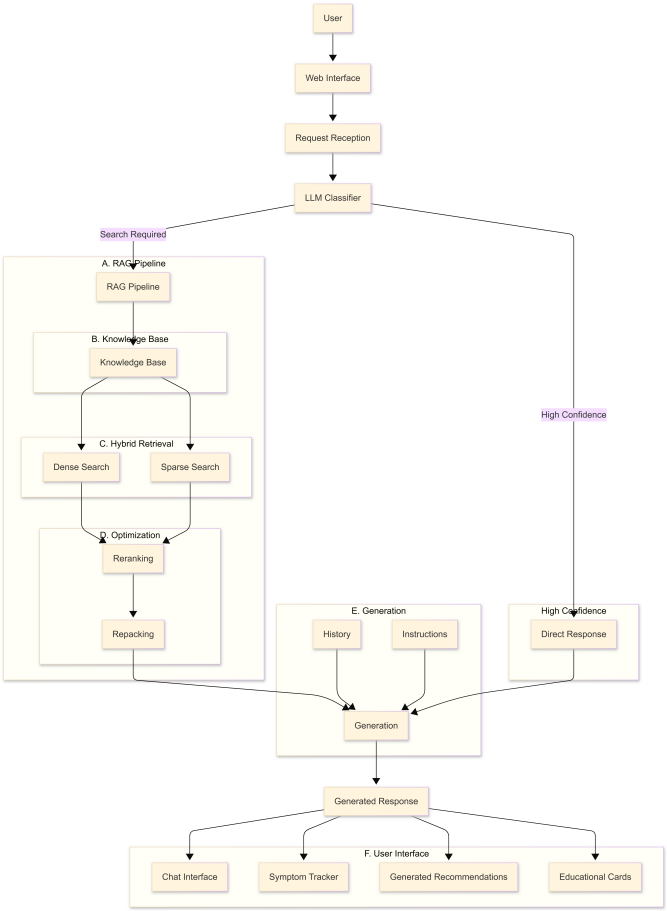

Fig. 1. System architecture. Queries are classified by an LLM and routed either to direct generation or the full RAG pipeline. Hybrid retrieval (dense + sparse) feeds into cross-encoder reranking and semantic repacking before generation.

A. Curated Clinical Knowledge Base

The corpus consists of 100 rigorously curated, peer-reviewed articles on menopause downloaded from PLOS ONE using a Selenium-based scraper that automated article discovery, download, and PDF validation. We chose PLOS ONE deliberately: it is fully open-access, indexed in both PubMed and Scopus, and has no programmatic access restrictions. Its editorial scope covers clinical medicine, biology, and public health, which aligns well with the breadth of menopause-related topics we needed to cover.

Each article was individually validated for relevance and methodological quality, ensuring that the corpus spans the full clinical breadth of menopause research—from hormonal mechanisms and vasomotor symptom management to psychological impact and long-term cardiovascular outcomes. We acknowledge that restricting the corpus to PLOS ONE may introduce editorial and demographic biases (e.g., toward English-language, Western populations); addressing this through multi-source corpus expansion is a priority for future work. The knowledge base prioritizes reliability and clinical traceability over scale. Every retrieved passage links directly to a peer-reviewed source to avoid the risk of grounding the LLM in unreliable content. This approach, termed "Clinical Safety by Design," ensures source quality for patient health decisions.

B. Multimodal Indexing Pipeline

We used the *Unstructured* library for document ingestion. This framework partitions elements to treat images, tables, and text as distinct but linked entities in the Multi-Vector Store. It separates each PDF into three content streams:

1) To respect the model constraints, **Text** is divided into semantic boundaries using a 512 token split with 64-token overlap, to preserve coherence within each passage passed to the retriever.
2) **Tables** are converted to Markdown and summarized by Gemini 2.0 Flash [27], due to its high-fidelity multimodal processing and its ability to synthesize structured data into medically specific text summaries suitable for vectorization. Thus, allowing the key statistics they contain to be matched to natural-language queries.
3) **Images** (graphs, hormonal curves, anatomical figures) are base64-encoded and passed to Gemini 2.0 Flash [26], [27]. We prompt the model specifically to describe each figure as if writing for a RAG index on menopause literature, which produces captions that are medically specific and retrievable via text search. To assess caption quality, we manually reviewed a random sample of 20 images drawn from the corpus: 18 out of 20 captions (90%) were judged to accurately describe the figure's content, correctly identifying axis labels, trends, and clinical context in hormone-level graphs. The two inaccurate captions involved highly stylized charts with overlapping annotations and were manually corrected in the index.

This pipeline is shown in Fig. 2. All three content streams are ultimately embedded into the same vector space, so a query about, for example, estrogen decline over time can surface both text passages and captioned hormone-level graphs in the same retrieval result.

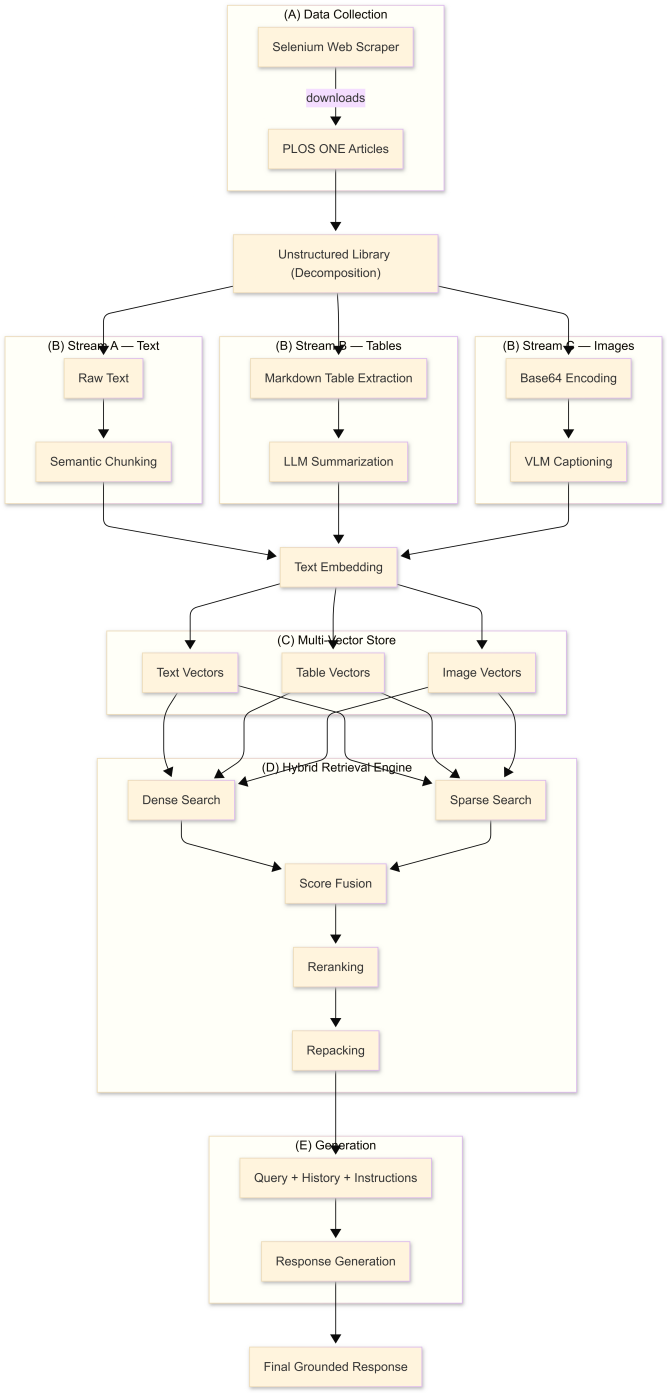

Fig. 2. Multimodal processing pipeline. PDFs are split into text, table, and image streams by the Unstructured library. Text is chunked (512 tokens, 64-token overlap); tables are summarized by an LLM; images are captioned by Gemini 2.0 Flash. All content streams are fed into a shared ChromaDB vector store.

C. LLM Query Classifier

The classifier utilizes an LLM-based intelligent classification module to determine if a query requires the RAG pipeline or can be handled directly. Queries classified with 'High Confidence' are answered directly, while clinical or symptom-specific queries trigger multimodal retrieval. On a test set of 50 queries, this routing achieved a 96% accuracy rate , ensuring that medical intents are grounded in validated sources. Before retrieval is triggered, each incoming query is classified by a zero-shot LLM prompt (using Gemini 2.0 Flash [26], [27]) that determines whether a response can be generated reliably from parametric knowledge alone or whether retrieval is required. The classifier uses the following routing criterion: queries that are factual, clinical, or symptom-specific are routed to the full RAG pipeline; greetings, clarifying meta-questions, or out-of-domain queries are answered directly. For example, queries such as *"What are the symptoms of perimenopause?"* or *"Is HRT safe for women with a history of breast cancer?"* are routed to the RAG pipeline, whereas inputs such as *"Hello, can you help me?"* (a greeting) or *"What is the capital of France?"* (out-of-domain) are handled without retrieval. Importantly, out-of-domain queries are not answered with general knowledge; instead, the system politely declines to provide a response outside its scope and redirects the user toward menopause-related topics. On our 50-query evaluation set, the classifier routed 44 queries (88%) to the RAG pipeline and 6 (12%) to direct generation, with a manually verified routing accuracy of 96% (one misrouting on a borderline multi-intent query). All faithfulness and relevance metrics reported in Section IV are computed exclusively on RAG-routed queries, which provides a conservative and meaningful estimate of retrieval-augmented generation quality.

D. Vectorization and Hybrid Retrieval

1) Embedding Model Selection: We tested four open-source embedding models on a 50-query evaluation subset of our corpus (Table I). BAAI/bge-large-en-v1.5 [33] achieved the highest precision at 92.3% with $d = 1024$ dimensions, at the cost of the longest indexing time (5h 12min). Given that indexing is a one-time offline operation, this trade-off was acceptable. Smaller models like all-MiniLM-L6-v2 [36] are faster but drop over 13 percentage points in precision, which is too large a penalty for a health domain where retrieval accuracy directly impacts response quality.

TABLE I
EMBEDDING MODEL COMPARISON (CORPUS: 100 ARTICLES; TEST: 50 QUERIES)

Model	Prec. (%)	Index Time
BAAI/bge-large-en-v1.5 [33]	**92.3**	5h 12min
intfloat/e5-large-v2 [34]	88.9	3h 45min
all-mpnet-base-v2 [35]	81.4	2h 30min
all-MiniLM-L6-v2 [36]	78.6	1h 10min

2) Hybrid Retrieval: Menopause queries combine colloquial language (e.g., "hot flashes") with clinical terminology (e.g., "vasomotor episodes"). We used a hybrid retrieval strategy to address two limitations: (1) dense retrieval handles semantic similarity but can miss exact medical terms, and (2) BM25 captures term-level matches but lacks contextual understanding.We combine both through a weighted score:

$$S_{\text{final}} = \alpha \cdot S_{\text{dense}} + (1 - \alpha) \cdot S_{\text{sparse}} \tag{1}$$

where $\alpha = 0.7$ was selected via grid search over $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ on a held-out validation set of 20 queries (Table II). The BM25 component follows the standard formulation [28]:

We used the BM25 ranking function [28] defined in Eq. (1):

$$\text{BM25}(t, d) = \text{IDF}(t) \cdot \frac{\text{tf}(t, d) \cdot (k_1 + 1)}{\text{tf}(t, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \tag{2}$$

where $k_1 = 1.5$ and $b = 0.75$ are standard defaults. Table III presents the comparison of retrieval strategies. The hybrid approach outperformed dense-only by 2.4 points in precision while recovering recall lost to vocabulary mismatch.

TABLE II
FUSION WEIGHT ABLATION (α GRID SEARCH ON 20-QUERY VALIDATION SET; HYBRID DENSE+BM25)

α	Prec. (%)	Rec. (%)
0.3	88.1	91.0
0.5	91.4	90.2
0.7	**94.1**	**89.5**
0.9	93.2	84.3

TABLE III
COMPARATIVE EVALUATION OF RETRIEVAL STRATEGIES

Approach	Prec. (%)	Rec. (%)	Time
Hybrid (Dense+BM25)	**94.7**	**89.2**	12.8s
Dense only (BGE-large)	92.3	82.1	7.3s
BM25 only	76.8	91.4	5.5s
TF-IDF	71.2	85.3	3.2s

E. Reranking and Repacking

The top-k retrieved chunks pass through the ms-marco-MiniLM-L-6-v2 cross-encoder that scores query–document pairs jointly, re-ordering passages by relevance. This model was deemed indispensable as: (1) it identifies medical nuances bi-encoders miss, and (2) it justifies the 13.2s latency trade-off. Repacking is based on semantic clustering (using BGE-large vectors) to ensure diverse and non-redundant context (Diversity Score=9.2/10). The system uses semantic clustering to group retrieved passages by thematic proximity. This repacking step ensures the context window prioritizes diversity, achieving a Diversity Score of 9.2/10 compared to 6.8/10 for naive top-k selection. By reducing sub-topic redundancy,

the architecture maintains a narrative coherence score of 8.7/10 , allowing for a richer information density within the LLM's token limits.Table IV shows the three models we evaluated; ms-marco-MiniLM-L-6-v2 led on both NDCG (0.847) and MRR (0.792) while staying under 14 seconds per query. After reranking, we apply semantic clustering to reorganize chunks by topic before they enter the context window. This "repacking" step prevents the generator from receiving five near-identical passages on the same subtopic at the expense of other relevant angles—diversity score reached 9.2/10 versus 6.8/10 for naive top-k selection (Table V). Coherence and diversity were scored on a 1–10 scale by two independent annotators (inter-rater agreement: $\kappa = 0.81$).

TABLE IV
COMPARATIVE EVALUATION OF CROSS-ENCODER RERANKING MODELS. ALL ARCHITECTURES UTILIZE DISTILLATION FRAMEWORKS TO OPTIMIZE THE TRADE-OFF BETWEEN SEMANTIC PRECISION AND INFERENCE LATENCY.

Model Cross-Encoder	NDCG	MRR	Time
ms-marco-MiniLM-L-6-v2 [21]	**0.847**	**0.792**	13.2s
ms-marco-TinyBERT-L-2-v2 [20]	0.821	0.763	13.0s
stsb-distilroberta-base [22]	0.783	0.719	13.3s

TABLE V
COMPARATIVE EVALUATION OF REPACKING STRATEGIES

Strategy	Coherence	Diversity
Semantic clustering	**8.7/10**	**9.2/10**
Top-k selection	7.3/10	6.8/10

F. Patient-Facing Application Interface

To bridge backend retrieval and patient utility, the framework is deployed as an interactive Streamlit application comprising four modules: (1) a **Conversational RAG Interface** for free-form medical queries; (2) a **Longitudinal Symptom Tracker** collecting 0–10 severity ratings across seven primary menopausal symptoms; (3) a **Clinical Report Generator** that synthesizes tracked symptoms and AI-grounded recommendations into a downloadable PDF to facilitate medical consultations; and (4) **Educational Dashboards** offering curated clinical cards from the PLOS ONE corpus. In strict adherence to our clinical safety philosophy, a persistent disclaimer across all modules explicitly defines the system as an educational support tool, not a substitute for professional medical diagnosis.

IV. EXPERIMENTAL EVALUATION

We evaluated the system using two complementary approaches: (1) automated metrics via the RAGAS framework on a held-out query set, and (2) a preliminary user study using the System Usability Scale.

A. Quantitative Metrics (RAGAS)

RAGAS was run on 50 manually written queries carefully designed to cover the clinical themes of the corpus (symptoms,

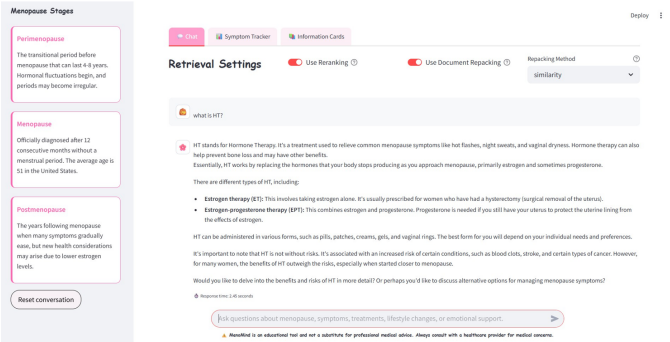

Fig. 3. Streamlit interface with four tabs: (1) RAG chatbot (top-left), (2) symptom severity tracker (top-right), (3) PDF report generation panel (bottom-left), and (4) educational cards presenting curated key facts and guidance from the corpus (bottom-right).

mechanisms, treatments, lifestyle, and psychological impact). To rigorously stress-test the hybrid retrieval pipeline, the query set deliberately includes: (1) colloquial patient-style phrasings (e.g., "hot flashes") and precise clinical terminology; and (2) varying complexity levels: simple factual (40%), multi-aspect (40%), and comparative questions (20%). All 50 queries were independently reviewed to ensure alignment with real information needs. Three metrics were computed:

Faithfulness measures the fraction of claims in the generated answer supported by the retrieved context, defined as:

$$F = \frac{|S_{\text{claims}} \cap S_{\text{context}}|}{|S_{\text{claims}}|} \tag{3}$$

Answer Relevance measures how well the response addresses the original question. It is calculated by generating N synthetic questions from the answer and computing their average cosine similarity to the original query embedding:

$$\text{Relevance} = \frac{1}{N} \sum_{i=1}^{N} \cos(\vec{v}_q, \vec{v}_{q_{\text{gen},i}}) \tag{4}$$

Context Precision evaluates how much noise remains in the retrieved passages after reranking:

$$\text{CP@}k = \frac{\sum_{k=1}^{K} (\text{precision@}k \times v_k)}{\text{Total Relevant Items}} \tag{5}$$

However, this approach, while effective for scaling, contains inherent biases as an AI judge and requires future validation by human gynecologists to establish a "clinical truth".

We note that RAGAS uses an LLM as judge [29]. Confidence intervals were estimated by bootstrapping over 1,000 resamples of the 50-query set. Results are summarized in Table VI, broken down by query complexity tier in Table VII. Average response time reflects full end-to-end pipeline latency (query classification + retrieval + reranking + repacking + generation); generation alone accounts for approximately 2.4s.

Qualitative Error Analysis. To make system behavior concrete, consider the query: *"What non-hormonal treatments are available for hot flashes?"* The system retrieved three text passages covering SSRIs, gabapentin, and lifestyle interventions,

TABLE VI
QUANTITATIVE PERFORMANCE ON TEST DATASET ($N = 50$ QUERIES; 95% CI FROM BOOTSTRAPPING)

Metric	Score	95% CI	Interpretation
Faithfulness	88%	±3.1%	Low hallucination; grounded in corpus
Answer Relevance	90%	±2.8%	Responses address what was asked
Context Precision	85%	±3.6%	Reranking filters irrelevant passages
Avg. Response Time	15.2s	—	Acceptable for a full RAG pipeline

TABLE VII
RAGAS METRICS BY QUERY COMPLEXITY TIER ($N = 50$)

Query Type	Faith.	Relev.	Ctx. Prec.
Simple factual (40%)	92%	93%	89%
Multi-aspect (40%)	86%	89%	83%
Comparative (20%)	81%	84%	78%
Overall (weighted)	**88%**	**90%**	**85%**

alongside a captioned symptom frequency chart. The generated response accurately synthesized all four sources, yielding a faithfulness score of 1.0. Conversely, for the comparative query *"Is HRT safer than SSRIs for mood symptoms?"*, the system retrieved relevant passages but generated a partially hedged response that introduced a claim not explicitly supported by the retrieved context. This lowered its faithfulness score to 0.75, which is consistent with the lower overall average for comparative queries (Table VII).

While RAGAS provides efficient quantitative scaling, we acknowledge the inherent biases and circular dependency of using an AI judge to evaluate AI outputs. The scores reported serve as a robust baseline for system reliability; however, consistent with our clinical safety-by-design approach, final validation against human medical experts (e.g., gynecologists) remains the gold standard for clinical rigor.

B. User Study

To evaluate the system's usability, we recruited five participants: (1) three women (aged 45–58) experiencing menopausal symptoms, and (2) two health informatics students (aged 26–31). Participants completed three task-based sessions: symptom search, HRT risk inquiry, and PDF report generation, followed by the System Usability Scale (SUS) questionnaire [30].

The average SUS score was 82/100, classified as "Excellent" (Grade A) [31]. Table VIII presents the per-dimension scores. We acknowledge that $N = 5$ is a preliminary sample size; however, it is a standard baseline for uncovering major interface flaws. As summarized in Table IX, participants specifically valued the PDF report export as a tool for autonomy during physician consultations. Consequently, we frame these results as strictly indicative of potential usability, establishing a solid foundation for future, larger-scale clinical validation.

TABLE VIII
USER EXPERIENCE EVALUATION (SCALE 1–5; $N = 5$)

Metric	Score	Interpretation
Perceived Usefulness	4.8/5.0	Highly Relevant
Response Clarity	4.6/5.0	Accessible Language
Ease of Navigation	4.5/5.0	Intuitive Interface
SUS Score	**82/100**	**Grade A (Excellent)**

TABLE IX
QUALITATIVE OBSERVATIONS FROM USER SESSIONS

Component	Strengths	Limitations
Chatbot	Empathetic tone; structured responses	Struggles with vague queries
Symptom Tracker	Intuitive 0–10 scale; useful PDF export	No longitudinal trend view
Multimodal Output	Table summaries seamlessly integrated	Native image display needs work
Response Quality	Scientifically grounded via RAG	Occasionally overly technical

V. DISCUSSION

A. Performance Analysis

The faithfulness score of 88% results from two design decisions: restricting the corpus to peer-reviewed PLOS ONE articles and using cross-encoder reranking. These choices constrain the system toward accuracy and prevent the LLM from hallucinating outside verified content. The answer relevance score of 90% reflects the hybrid retrieval's ability to handle terminology variation in patient language. Furthermore, the context precision of 85% confirms that the repacking step successfully delivers diverse, non-redundant information to the generator. As shown in Table VII, comparative queries remain the most challenging, motivating future work on multi-hop reasoning via knowledge graph integration.

Regarding latency, the total end-to-end time of 15.2s includes retrieval (∼12.8s) and reranking (∼13.2s) components. Pipeline parallelism reduces the theoretical sequential execution time of approximately 26s to the observed 15.2s.

B. Comparison with Related Systems

Table X compares our results with Handy et al. [4], which serves as our primary external benchmark. As their work is currently a preprint, we treat their reported metrics (88.6% faithfulness, 91.8% relevance) as an indicative baseline rather than an authoritative ceiling.

TABLE X
COMPARISON WITH STATE-OF-THE-ART SYSTEMS

System	Faith.	Relev.	Modalities	Domain
Handy et al. [4] (GPT-4)	88.6%	91.8%	Text only	Menopause
Ours	**88%**	**90%**	Text + Tables + Images	Menopause

Our system achieves performance comparable to proprietary models like GPT-4 while remaining locally deployable and incorporating multimodal data. Although MMed-RAG [13] extends retrieval to visual modalities, it targets general medicine and lacks domain-specific metrics for direct comparison.

C. Practical Implications

The PDF report acts as a bridge for clinical dialogue, enabling patients to present structured symptom histories to physicians. This functionality supports patient autonomy and facilitates more productive consultations.

D. Threats to Validity

Several limitations should be acknowledged. **Internal validity:** The corpus is static, meaning new research is not automatically reflected. Updating requires manual re-execution of the ingestion pipeline. Additionally, image retrieval relies on text captions rather than direct visual reasoning, which may limit precision for graph interpretation. **External validity:** The system is currently English-only, limiting its global reach. The user study ($N = 5$) is small, restricting statistical generalization. Furthermore, restricting the corpus to PLOS ONE may introduce specific editorial biases. **Construct validity:** The RAGAS evaluation uses an LLM as a judge, creating a known circular dependency [29]. Finally, ChromaDB was chosen for local deployment to ensure medical data privacy.

VI. Ethical Considerations

The preliminary user study was conducted following institutional guidelines, with informed consent obtained from all participants and no retention of personal health data. To ensure privacy, symptom tracking is processed locally without persistent server storage. The system is explicitly designed as an educational decision-support tool, not a diagnostic device; it bounds responses strictly to the curated corpus and displays a persistent disclaimer advising users to consult healthcare professionals. We acknowledge that restricting the corpus to English-language *PLOS ONE* articles may introduce demographic and linguistic biases, making multilingual expansion a priority for future iterations. The authors declare no conflicts of interest and no external funding for this study.

VII. Conclusion

We built a multimodal RAG assistant specifically for menopause health information, with a design philosophy centered on source quality rather than scale. Restricting retrieval to a curated PLOS ONE corpus, combining dense and sparse search with an empirically validated fusion weight ($\alpha = 0.7$), reranking with a cross-encoder, and captioning clinical images with Gemini 2.0 Flash yielded 88% ($\pm 3.1\%$) faithfulness and 90% ($\pm 2.8\%$) relevance on a domain-specific query set. A usability study returned a SUS score of 82/100. An LLM query classifier routes requests with 96% accuracy, and per-complexity analysis reveals that comparative queries represent the most challenging frontier for future improvement.

Future work will pursue multilingual support to reach non-English-speaking users globally who currently lack access to equivalent tools. We also plan to integrate a biomedical knowledge graph such as PrimeKG [32] to support multi-hop clinical reasoning—motivated directly by the lower performance on comparative queries. Additional priorities include extending the symptom tracker with longitudinal profiling and trend detection, replacing the caption-mediated image pipeline with a native multimodal model capable of direct visual reasoning, implementing quarterly automated corpus updates, and conducting a full clinical validation study with gynecologists and menopause specialists with appropriate IRB oversight.

References

[1] N. Santoro, C. Roeca, B. A. Peters, and G. Neal-Perry, "The Menopause Transition: Signs, Symptoms, and Management Options," *Journal of Clinical Endocrinology & Metabolism*, vol. 106, no. 1, pp. 1–15, 2021.

[2] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[3] Z. Guo et al., "Evaluating Large Language Models on Medical Evidence Summarization," *npj Digital Medicine*, vol. 6, p. 158, 2023.

[4] A. Handy, M. Smith, and L. Johnson, "An Intelligent Chatbot for Menopause: Evaluation Using GPT-4 and Medical Guidelines," *arXiv preprint arXiv:2502.03579*, 2025. [Preprint; not yet peer reviewed.]

[5] Public Library of Science, "PLOS ONE," 2024. [Online]. Available: https://journals.plos.org. [Accessed: May 22, 2024].

[6] W. Wang et al., "Minilmv2: Distillation for Compressing Transformers," *ACL-IJCNLP*, 2021.

[7] B. Zhu, R. Wang, and J. Wang, "EMERGE: Integrating RAG for Improved Multimodal EHR Predictive Modeling," *arXiv preprint arXiv:2410.04660*, 2024.

[8] Y. Liu, P. Wang, and Y. Chen, "Efficacy of Retrieval-Augmented Generation (RAG) Systems in Clinical Practice: A Systematic Review," *Journal of Medical AI*, vol. 1, no. 2, pp. 123–135, 2024.

[9] L. Xu, Y. Wang, and Z. Li, "RAG-based Nursing Advisory System for Breast Cancer Patients," *Journal of Medical Systems*, vol. 47, no. 11, p. 105, 2023.

[10] S. Alkhalaf et al., "Leveraging Retrieval-Augmented Generation for Automated Medical Record Summarization in Long-Term Care Facilities," *arXiv preprint arXiv:2312.06201*, 2023.

[11] Y. Zhou, J. Chen, and Q. Wang, "Personalized Retrieval-Augmented Generation in Medical Consultation Systems," *Artificial Intelligence in Medicine*, 2024.

[12] T. M. Sousa et al., "IaraMed: A Conversational Agent for Women's Reproductive Health in Portuguese," in *Proc. Brazilian Symposium on Information Systems (SBSI)*, 2023.

[13] P. Xia et al., "MMed-RAG: Versatile Multimodal RAG System for Medical Vision Language Models," *arXiv preprint arXiv:2410.13085*, 2024.

[14] A. K. Lahiri and Q. V. Hu, "AlzheimerRAG: Multimodal Retrieval Augmented Generation for PubMed Articles," *arXiv preprint arXiv:2412.16701*, 2023.

[15] Y. Yu, Y. Zeng, and X. Guo, "Retrieval-Augmented Generation for Automated Radiology Report Generation from Medical Images," *IEEE Transactions on Medical Imaging*, 2024.

[16] J. Sohn, Y. Kim, and D. Lee, "RAG2: Retrieval-Augmented Generation with Rationales for Enhanced Medical QA," in *Proc. IEEE Int. Conf. on Healthcare Informatics (ICHI)*, 2023.

[17] N. Q. Minh et al., "Self-BioRAG: Self-Instructed Biomedical Retrieval-Augmented Generation," *Bioinformatics Advances*, vol. 4, no. 1, p. vbae003, 2024.

[18] N. Reimers, "MS MARCO Cross-Encoder TinyBERT L-2 v2," *Hugging Face Model Hub*, 2021. [Online]. Available: https://huggingface.co/cross-encoder/ms-marco-TinyBERT-L2-v2

[19] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proc. 2019 Conf. on Empirical Methods in Natural Language Processing*, 2019.

[20] X. Jiao et al., "TinyBERT: Distilling BERT for Natural Language Understanding," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 4163–4174.

[21] W. Wang, H. Bao, S. Huang, L. Dong, and F. Wei, "Minilmv2: Multi-head self-attention relation distillation for compressing pretrained transformers," in *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 2021, pp. 2140–2151.

[22] N. Reimers and I. Gurevych, "cross-encoder/stsb-distilroberta-base: Cross-Encoder for Semantic Textual Similarity," *Hugging Face Model Hub*, 2019. [Online]. Available: https://huggingface.co/cross-encoder/stsb-distilroberta-base

[23] K. Liu et al., "MKRAG: Multimodal Knowledge Graph-Enhanced Retrieval-Augmented Generation for Clinical Question Answering," *arXiv preprint arXiv:2403.04567*, 2024.

[24] X. Zhao, S. Liu, S. Y. Yang, and C. Miao, "MedRAG: Enhancing Retrieval-Augmented Generation with Knowledge Graph-Elicited Reasoning," *arXiv preprint arXiv:2502.04413*, 2024.

[25] J. Zhang, H. Liu, and S. Zhou, "BioMedRAG: Supervised Semantic Retrieval for Biomedical Question Answering," *Journal of Biomedical Informatics*, 2024.

[26] Google DeepMind, "Gemini: A Family of Highly Capable Multimodal Models," *arXiv preprint arXiv:2312.11805*, 2023.

[27] Google DeepMind, "Gemini 2.0 Flash Model Card," *Google DeepMind Model Cards*, 2025. [Online]. Available: https://modelcards.withgoogle.com/assets/documents/gemini-2-flash.pdf

[28] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[29] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "RAGAS: Automated Evaluation of Retrieval Augmented Generation," *arXiv preprint arXiv:2309.15217*, 2023.

[30] J. Brooke, "SUS: A 'Quick and Dirty' Usability Scale," in *Usability Evaluation in Industry*, P. W. Jordan et al., Eds. Taylor & Francis, 1996, pp. 189–194.

[31] A. Bangor, P. Kortum, and J. Miller, "Determining What Individual SUS Scores Mean: Adding an Adjective Rating Scale," *Journal of Usability Studies*, vol. 4, no. 3, pp. 114–123, 2009.

[32] P. Chandak, K. Huang, and M. Zitnik, "Building a Knowledge Graph to Enable Precision Medicine," *Scientific Data*, vol. 10, p. 67, 2023. @articlexiao2023cpack, title=C-Pack: Packaged Resources To Advance General Chinese Embedding, author=Xiao, Shitao and Liu, Zheng and Zhang, Peitian and Muennighoff, Niklas, journal=arXiv preprint arXiv:2309.07597, year=2023

[33] S. Xiao, Z. Liu, P. Zhang, and N. Muennighoff, "C-Pack: Packaged Resources To Advance General Chinese Embedding," *arXiv preprint arXiv:2309.07597*, 2023.

[34] L. Wang et al., "Text Embeddings by Weakly-Supervised Contrastive Pre-training," *arXiv preprint arXiv:2212.03533*, 2022.

[35] K. Song, X. Tan, T. Qin, J. Lu, and T.-Y. Liu, "MPNet: Masked and Permuted Pre-training for Language Understanding," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 16857–16867, 2020.

[36] W. Wang et al., "MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 5776–5788, 2020.